EE 451: Communications Systems — Midterm Exam 2 Formula Sheet

FM/PM Formulas:

Name	Formula
FM instantaneous frequency	$f_i(t) = f_c + k_f \cdot m(t)$
FM frequency deviation	$\Delta f = k_f \cdot \max\lvert m(t)\rvert$
FM modulation index	$\beta = \Delta f / f_m$
Carson's rule (FM bandwidth)	$B_{\text{FM}} = 2(\Delta f + f_m) = 2f_m(\beta + 1)$
NBFM criterion	$\beta \ll 1$ (typically $\beta < 0.3$)
PM instantaneous phase	$\theta_i(t) = 2\pi f_c t + k_p \cdot m(t)$
PM phase deviation	$\Delta\phi = k_p \cdot \max\lvert m(t)\rvert$
PM instantaneous frequency	$f_i(t) = f_c + \frac{k_p}{2\pi}\frac{dm(t)}{dt}$
PM frequency deviation (sinusoidal)	$\Delta f_{\text{PM}} = k_p \cdot A_m \cdot f_m$

Digital Modulation Formulas:

Name	Formula
Bit duration	$T_b = 1/R_b$
FSK modulation index	$h = \Delta f \cdot T_b$
FSK orthogonality (coherent)	$\Delta f_{\min} = 1/(2T_b) = R_b/2$
MSK (Minimum Shift Keying)	$h = 0.5$, so $\Delta f = R_b/2$
FSK bandwidth approximation	$B_{\text{FSK}} \approx 2\Delta f + R_b$
Null-to-null bandwidth (BPSK/ASK)	$B = 2R_b$
M-ary bits per symbol	$k = \log_2(M)$
Symbol rate	$R_s = R_b/k$
QPSK null-to-null bandwidth	$B_{\text{QPSK}} = 2R_s = R_b$

Name	Formula
Spectral efficiency	$\eta = R_b/B$ (bits/s/Hz)

Noise Formulas:

Name	Formula
Thermal noise power	$P_n = kTB$ where $k = 1.38 \times 10^{-23}$ J/K
SNR (dB)	$\mathrm{SNR} = 10\log_{10}(P_s/P_n)$
Noise figure (linear)	$F = \mathrm{SNR}_{\mathrm{in}}/\mathrm{SNR}_{\mathrm{out}}$
Noise figure (dB)	$NF = 10\log_{10}(F)$
Equivalent noise temperature	$T_e = (F-1)\cdot T_0$
System noise temperature	$T_{\mathrm{sys}} = T_{\mathrm{ant}} + T_e$
Friis cascaded noise figure	$F_{\mathrm{total}} = F_1 + \frac{F_2-1}{G_1} + \frac{F_3-1}{G_1G_2} + \cdots$
Noise power (dBm)	$N = -174 + 10\log_{10}(B)$ where B is in Hz

Link Budget Formulas:

Name	Formula
EIRP	$\mathrm{EIRP} = P_t \cdot G_t$ (linear) or $P_t + G_t$ (dB)
Friis transmission equation	$P_r = P_tG_tG_r\left(\frac{\lambda}{4\pi d}\right)^2$
Free-space path loss (dB)	$L_p = 20\log_{10}(d) + 20\log_{10}(f) + 32.45$
(units for path loss formula)	d in km, f in MHz
Link margin	$\mathrm{Margin} = \mathrm{SNR} - \mathrm{SNR}_{\mathrm{req}}$

Useful Identities:

Name	Formula
dB conversion	$X_{\text{dB}} = 10\log_{10}(X)$
dBm to watts	$P_W = 10^{(P_{\text{dBm}}-30)/10}$
Watts to dBm	$P_{\text{dBm}} = 10\log_{10}(P_W) + 30$
Euler's formula	$e^{j\theta} = \cos\theta + j\sin\theta$

Reference: $T_0 = 290$ K (standard reference temperature). $k = 1.38 \times 10^{-23}$ J/K (Boltzmann's constant).